1. Definitions.

① Let A be a dg Q-alg. and M a dg A-mod.

Def: A Q-lin. derivation is a Q-lin. map $\theta: A \to B$ of degree $|\theta|$ such that: $\forall\, x, y \in A$

$$\theta(xy) = \theta(x)y + (-1)^{|\theta||x|} x\theta(y).$$

θ is a chain derivation if θ is a chain map $A \to \Sigma^{|\theta|} B$.

Def: $\mathrm{Der}_Q(A, M)$ is the set of all derivations;
$\mathrm{Der}_Q(A, M)_n$ is the set of deg. n derivations.

with $\begin{cases} \text{differential: } \partial(\theta) = \partial_M \theta - (-1)^{|\theta|}\theta\partial_A \\ A\text{-mod. structure: } a\cdot\theta(x) = a\theta(x). \end{cases}$

② Def: One can show that in cat. of dg A-mod., $\mathrm{Der}_Q(A, -)$ is representable i.e.

$$\mathrm{Der}_Q(A, M) \cong \mathrm{Hom}_A(\Omega_{A|Q}, M)$$

$\Omega_{A|Q}$ is the dg A-mod. of Kahler differentials.

For any derivation $\theta: A \to M$, $\exists\, d_\theta : \Omega_{A|Q} \to B$ s.t.:

$$\begin{array}{ccc} A & \xrightarrow{\theta} & M \\ {\scriptstyle d}\downarrow & \nearrow_{d_\theta} & \\ \Omega_{A|Q} & & \end{array}$$

fixed derivation

if θ chain deriv., then d_θ a dg A-mod. map.

Rmk: In nondg case:

$$\Omega_{A|Q} := \bigoplus_{a\in A} A\cdot da \,/\, (d(a+b)-da-db,\ d(q),\ d(ab)-a\,db-da\,b)$$

($d(a+b)-da-db$: Q-lin.; $d(ab)-a\,db-da\,b$: deriv.)

Similarly in dg case, $\Omega^{\natural}_{A|Q}$ is free A-mod.

2. Properties when $A = Q[X]$.

① Lem: Let $\theta: Q[X] \to Q[X]$ be a deriv. of deg. ≤ 0 s.t. $\theta(X_{\ge 0}) \subset (X_{\ge 0})$

$\begin{cases} \text{(i) } \theta((X_{\ge 0})^n) \subset (X_{\ge 0})^n \\ \text{(ii) } \theta \text{ induces } \theta^{(n)}: k[X_{\ge n}] \to k[X_{\ge n}]. \end{cases}$

Pf: (i) By induction & deriv., $\theta(X^n) \subset X^n$.

(ii) Since θ deg. ≤ 0, $\theta: X_{<n} \to X_{<n}$.

Thus $\theta^{(n)}: k[X]/X_{<n} \to k[X]/X_{<n}$ well-def.

② Key properties.

(i) $\Omega^{\natural}_{A|Q}$ is free A-mod. on $dX_i \xleftrightarrow{\text{deg. preserving bijection}} X_i$

(ii) • $\Omega_{A|Q} \to R \otimes_A \Omega_{A|Q}$ is a quasiisom.

(Note that $A \xrightarrow{\sim} R$ and $\Omega_{A|Q}$ cofree).

- $R\otimes_A \Omega_{A|Q}$ is minimal complex of free R-mod.

$$R\otimes_A \Omega_{A|Q} \cong \bigoplus_{i\geq 1} R\cdot dX_i. \qquad \text{(by tensoring)}.$$

- As an R-complex,

$$R\otimes_A \Omega_{A|Q} \cong R[X]/((X^2)+R) \cong RX$$

(iii) ✡ $H_1(\Omega_{A|Q}) \cong H_1(R\otimes_A \Omega_{A|Q}) \cong I/I^2$.

(iv) $\mathrm{Der}_Q(Q[X], -) \cong \mathrm{Hom}_{Q[X]}(\Omega_{Q[X]|Q}, -) \cong \mathrm{Hom}_R(R\otimes_{Q[X]} \Omega_{Q[X]|Q}, -)$

↓ Hom tensor adjunction ⇒ basis change.

Thus $\mathrm{Der}_Q(Q[X], -) \cong \mathrm{Hom}_R(RX, -)$.

Rmk: Another interpretation: deriv. on $Q[X]$ is determined by $\theta(x)$ for $x\in X$, by def. of deriv.

Thus: $\mathrm{Der}_Q(A, M) \xleftrightarrow{\cong} \prod_{x\in X} M$

$\theta: x_i \mapsto m_i \quad \longleftrightarrow \quad (m_i)_{x_i\in X}$

③ LES.

Since $\Omega_{A|Q}$ semifree when $A=Q[X]$, $\mathrm{Hom}_A(\Omega_{A|Q}, -)$ is exact:

$$0 \to U \to V \to W \to 0 \qquad \text{exact}$$

$$\Rightarrow \cdots \to H_n(\mathrm{Der}_R(A, U)) \to H_n(\mathrm{Der}_R(A, V)) \to \cdots$$

is an induced LES.

④ e.g.

$Q=k[[x,y,z]]$, $I=(xy, zx, yz)$, $R=Q/I$. Compute $R\otimes_{Q[x]} Q_{Q[x]|Q}$.

The complex is:

$$\cdots \to RX_3 \to RX_2 \to RX_1 \to 0$$

with differentials ∂ induced by $Q[x]$:

i.e. the linear part of $\partial_{Q[x]}$ ☆☆.

3. Thm: $\exists$ iso. of graded v.s.:

$$Der_Q(Q[x], k) \cong \pi^*(R).$$

Pf:

- $\theta: Q[x] \to k$ is det. by $X \to k$, and since k is in deg. 0, by degree reasons, only det. by $Xd \to k$, $d=\deg(\theta)$.
- Thus θ is det. by k-linear $kX_d \to k$, by def. dt. of $\pi^{d+1}(R)$ $=(\Sigma k X_d)^\vee$.

Thus $Der_Q(Q[x], k)_d \cong \pi^{d+1}(R)$, as a whole a graded v.s. isom.

Rmk: See 10.2.1 in AVR for another formulation of $\pi^*(R)$.

4. Derivation $Q[X] \to Q[X]$ associated to $z \in \pi^n(R)$.

① Construction: Given $z \in \pi^n(R)$, $z : X_{n-1} \to k$

induces Q-lin. map $QX_{n-1} \to k$

lifts to $\tilde{z} : QX_{n-1} \to Q$.

- Define: $\frac{d}{dz} : Q[X] \to Q[X]$

$$\frac{d}{dz}(x) = \begin{cases} \tilde{z}(x) & \text{if } x \in X_{n-1} \\ 0 & \text{if } x \in X_l \text{ for } l \neq n-1 \end{cases}$$

& extend by Leibniz.

- Chain derivation: $\theta_z := [\partial, \frac{d}{dz}]$.

② $\begin{cases} \theta_z : Q[X] \to Q[X] \text{ chain derivation} \\ \text{im}(\theta_z) \subset (X). \end{cases}$

pf:

- By applying ∂ and $\partial^2 = 0$, we can show θ_z chain map.
- For $x \in (X)$, WTS:

$$\partial \circ \frac{d}{dz}(x) - (-1)^{|x|-1} \frac{d}{dz} \circ \partial(x) \in (X).$$

Note that $\frac{d}{dz}(x) \in Q \Rightarrow \partial \circ \frac{d}{dz}(x) = 0$.

☆ We have: $\partial(X) \subset (X)^2$, and $\frac{d}{dz}(X)^2 \subset (X)$ since $\frac{d}{dz}$ a deriv.

$\Rightarrow \theta_z(X) \subset (X)$.

But then $\theta_z(Q[X]) \subset (X)$.